

Nicole Zhou

Redmond, WA 98052

(425)-591-0968 • nz278@cornell.edu

<https://zhounicole.com/>

EDUCATION

Cornell University, College of Engineering , Ithaca, NY	Expected May 2029
Intended Bachelor of Science in Electrical and Computer Engineering and Minor in Physics	GPA: 3.96
Coursework: Digital Logic, Embedded Systems, Data Science, Nanoscience and Nanoengineering	
Interlake High School , Bellevue, WA	June 2025
GPA: 4.0/4.0, IB: 42/45, SAT: 1600, AP Scholar with Distinction, U.S. Presidential Scholar Candidate	

TECHNICAL EXPERIENCE

Cornell Autonomous Sailboat Project Team	Ithaca, NY
Navigation Electrical Subteam	October 2025 – Present
- Prototyping an embedded remote control system for selective activation of onboard electronics - Using CAD to design a custom PCB for power distribution and physical panel to mount components	
Computer Systems Laboratory	Ithaca, NY
Napp Lab Embedded Systems Researcher	December 2025 – Present
- Developing autonomous construction robot that operates on decentralized coordination algorithms - Implementing embedded communication protocols based on custom motor driver library	
Center for Integrated Electronic and Biological Organisms	Los Angeles, CA
Computational Modeling Intern at Morsut Lab	June – August 2026
- Engineered synthetic genetic circuits and extended simulation framework for tissue patterning and morphogenesis - Developed quantitative FFT and autocorrelation analysis tools to characterize stem cell Turing patterns	
Fred Hutchinson Cancer Center	Seattle, WA
Machine Learning Intern at Dey Lab	September 2024 – June 2025
- Built and optimized a deep neural network model that integrated various feature-reduction techniques with multi-omic data to predict CRC risk in an Alaskan Native population with up to 80% accuracy - Performed rigorous checks on performance through permutation testing and class imbalance analysis	
MIT Beaver Works Summer Institute	Boston, MA
E-Textiles and Wearable Technology Program	June – August 2024
- Created a pulse oximeter device that integrated soft electronics into a custom textile sleeve for real-time monitoring of health metrics and presented to over 2,000 people both virtually and in-person	

SKILLS AND INTERESTS

Programs: Java, Python, C++, Verilog, Ubuntu, ROS2, Fusion 360, OnShape, KiCad, Altium, Git

Campus Involvement: Cornell Maker Club, IEEE at Cornell, Outdoor Odyssey

High School Extracurriculars: Jazz Ensemble, Club and Varsity Tennis, Red Cross

PREVIOUS BACKGROUND

HOSA , Bellevue, WA, Top Ten International Competitor	September 2021 – June 2024
Venture Academy Tutoring , Remote, English and Chinese Tutor	September 2022 – June 2024
Newport Hills Swim and Tennis Club , Newport, WA, Tennis Coach	June – August 2023